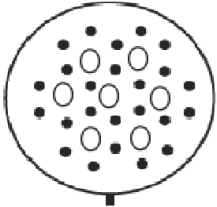


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		 Dots showing myosin filaments in correct position and an organised array of actin around the myosin		1		1		
		(ii)		• The actin {moves / slides} over the myosin / actin pulled by myosin (1) • Causing the H band to {shorten / get narrower} (1) • The myosin filaments do not shorten/change length (during contraction) / A band {shows / is} length of myosin filaments (1) • So the A band remains the same length (1)	2	2		4		
		(iii)		57.14 / 57.1 / 57 = 2 marks Award 1 mark for sight of (3.5-1.5) / 3.5 x 100		2		2	2	
		(iv)		Structure of actin: Any one (x1) of: {Actin / it} has myosin <u>binding sites</u> (1) Contraction: Calcium ions bind to troponin causing {troponin / tropomyosin} to {move / change shape} (1) Cross bridges are unable to form at rest: Any one (x1) of: Binding site is {covered / blocked} by tropomyosin when relaxed (1) No calcium ions bind to troponin (1) tropomyosin blocks myosin heads from forming cross bridges. (1)	2			2		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		Structure – high proportion of collagen (1) Function Any one (x1) from smooth to reduce friction (1) {Elastic / compressible} as a shock absorber (1) Hard to keep trachea open (1) Flexible to allow movement of ribcage (1)	1	1		2		
		(ii)		Auto-immune disease / immune system recognises proteins in the joint as antigens / Immune system attacks {bones / cartilage in joints} (1) Any 2 for one mark of: Cartilage / hyaline, bone synovial membrane / tissue	1	1		2		
	(c)	(i)		Any two (x1) from: {Thinner layer / less} compact bone / ORA (1) Spongy bone has become {less dense / have larger holes} / less bone mass (1) Less hydroxy apatite (1)		2		2		
		(ii)		Any three (x1) from: Treatment decreases the fracture rate compared to {group B / control} (1) Treatment does not decrease the fracture rate to the level in {group A / healthy} (1) Treatment increases the bone strength compared to {group B / control} (1) {group C / experimental} bone strength is not as great as {group A / healthy} (1)			3	3		3
		(iii)		Increased number of osteoblasts to rebuild bones (1) Increases bone density / strength (1)			2	2		
				Question 8 total	6	9	5	20	2	3